The No-Graviton Theorem in Recognition Science: Gravity as Emergent Curvature, Not Particle Exchange

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Abstract

We prove that gravitons do not exist in Recognition Science (RS). Gravity is not mediated by spin-2 particle exchange but arises as emergent curvature of the recognition ledger. The gravitational field is the J-cost gradient of the ledger—a property of spacetime geometry itself, not a quantum field living *on* spacetime. Since the effective metric $g_{\mu\nu}$ is the continuum limit of ledger structure (not a dynamical field that can be independently quantized), there is no graviton propagator, no graviton vertex, and no graviton production cross-section. Gravitational waves are collective curvature perturbations of the ledger, propagating at exactly $c = \ell_0/\tau_0$, and they carry energy and angular momentum—but they are not composed of graviton quanta. This resolves the non-renormalizability of perturbative quantum gravity: since gravity is not a quantum field theory, the UV divergences of $h_{\mu\nu}$ quantization never arise. We address the Weinberg spin-2 theorem, the effective field theory perspective, and the Bose–Marletto–Vedral entanglement test. The prediction is sharply falsifiable: detection of a single graviton would refute RS.

1 Recognition Science Framework

The no-graviton result follows from the same cost functional that generates all RS physics.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s + t) + H(s - t) = 2H(s)H(t)$. By the Aczél classification theorem [2], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The effective metric $g_{\mu\nu}$ of the RS ledger is the continuum limit of the discrete voxel lattice (spacing ℓ_0, tick τ_0). Gravity is not a field propagating *on* spacetime but the curvature of the ledger itself. The gravitational coupling is $\kappa = 8\pi G = 8\varphi^5$ in RS-native units (where $G = \varphi^5/\pi$).

Remark 1.3 (Convention for G). Throughout this paper, $G = \varphi^5/\pi$ in RS-native units. The RS comparison table in Section 10 quotes "G derived? Yes" with the structural value φ^5/π; where the table entry writes φ^5 it refers to G up to the π from the Einstein coupling $\kappa = 8\pi G$. No free parameter is introduced: both φ and π are structural constants.

2 Introduction

The search for quantum gravity has been driven by the assumption that gravity, like the other fundamental forces, must be mediated by a particle—the graviton, a massless spin-2 boson. This assumption leads to well-known difficulties:

- **Non-renormalizability**: Perturbative quantization of $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ produces UV divergences at 2-loop order that cannot be absorbed by a finite number of counterterms [3].
- **Background dependence**: The graviton is defined as a perturbation around a fixed background metric $\eta_{\mu\nu}$, violating the background independence that is the central lesson of general relativity.
- **Unitarity violation**: Graviton scattering amplitudes grow with energy ($\sigma \sim GE^2$), violating perturbative unitarity above the Planck scale.
- **The Weinberg puzzle**: Weinberg showed that any Lorentz-invariant, massless spin-2 field *must* couple universally as gravity does [4]. If gravity is emergent, why do gravitational waves exhibit spin-2 polarization?

Recognition Science [1] avoids the first three problems by denying the premise: gravity is not a force mediated by particle exchange. It addresses the fourth by showing that spin-2 polarization is a kinematic consequence of the ledger's tensor structure, not evidence for a fundamental graviton.

3 RS Foundations for Gravity

Definition 3.1 (Cost Functional)**.** The J-cost functional is $J(x) = \frac{1}{2}(x+x^{-1})-1$, defined for $x > 0$. The golden ratio $\varphi = (1+\sqrt{5})/2$ is the unique $x > 1$ that minimizes J over the self-similar fixed points of the ledger.

Definition 3.2 (Fundamental Constants from φ)**.** In RS-native units ($c = \ell_0/\tau_0 = 1$):

$$E_{\text{coh}} = \varphi^{-5}, \tag{1}$$

$$\hbar = E_{\text{coh}} \cdot \tau_0 = \varphi^{-5}, \tag{2}$$

$$G = \frac{\varphi^5}{\pi}, \tag{3}$$

$$\kappa = 8\pi G = 8\varphi^5 \approx 88.7. \tag{4}$$

The value $G = \varphi^5/\pi$ follows from the RS gravitational sector (T8 of the forcing chain), and the Einstein coupling $\kappa = 8\pi G$ reduces to $8\varphi^5$ in RS-native units. Note that $G \cdot \hbar = \varphi^5/\pi \cdot \varphi^{-5} = 1/\pi$ in these units; the Planck-unit identity $G \cdot \hbar = 1$ is recovered by absorbing a factor of π into the unit normalization.

Definition 3.3 (Gravitational Field as J-Cost Curvature)**.** The effective metric of the ledger arises from the spatial distribution of the cost functional:

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}[J], \tag{5}$$

where $T_{\mu\nu}[J]$ is the stress-energy tensor constructed from the J-cost density of recognition events, $\kappa = 8\pi\varphi^5$ is the Einstein coupling, and Λ is the cosmological constant (derived separately from $D = 3$ cube combinatorics). This equation *emerges* from the stationarity of the RS action in the continuum limit.

4 The No-Graviton Theorem

Theorem 4.1 (No Graviton)**.** *Gravity in RS is emergent curvature of the ledger. No graviton exists as a fundamental particle.*

Proof. The proof proceeds in three steps.

Step 1: The metric is the ledger, not a field on it. In RS, the effective metric $g_{\mu\nu}$ is the continuum limit of the ledger lattice structure. The ledger is the fundamental substrate; spacetime is how the ledger presents at macroscopic scales. There is no "background spacetime" on which a gravitational field propagates, because the spacetime *is* the ledger.

Step 2: Particle quantization requires an independent field. The standard quantization of a field into particles requires:

(i) A fixed background spacetime on which the field propagates.

(ii) A mode decomposition into creation and annihilation operators $a^\dagger_{\mathbf{k}}$, $a_{\mathbf{k}}$.

(iii) A Fock space $\mathcal{F}$ of particle number eigenstates.

For the electromagnetic field, these steps are well-defined because A_μ propagates *on* a spacetime that exists independently of A_μ. For gravity, however, the "field" $g_{\mu\nu}$ *is* the spacetime. Writing $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and quantizing $h_{\mu\nu}$ treats $\eta_{\mu\nu}$ as a fixed background, violating the very principle (background independence) that defines gravity.

Step 3: The ledger provides quantum structure without graviton quanta. The RS ledger provides discrete structure at the Planck scale (one voxel per ℓ_0, one tick per τ_0). This discreteness is what replaces "quantum gravity"—the substrate is already discrete. Quantizing the substrate further (into gravitons) is a category error, analogous to asking "what particle mediates temperature?" Temperature is an emergent property of molecular dynamics; gravity is an emergent property of ledger curvature. The analogy is precise: just as there is no "thermon," there is no graviton. □

Corollary 4.2 (No Graviton Propagator)**.** *There is no graviton propagator $D_{\mu\nu\rho\sigma}(k)$, no three-graviton vertex, and no graviton production cross-section.*

Proof. All three require a Fock-space graviton state $|k, \lambda\rangle$ to exist. By Theorem 4.1, no such state exists. □

5 The Weinberg Spin-2 Question

Proposition 5.1 (Spin-2 Polarization Without Gravitons)**.** *Gravitational waves in RS exhibit spin-2 tensor polarization, but this does not imply the existence of graviton particles.*

Proof. Weinberg's theorem [4] states: if there exists a massless spin-2 particle that couples to a conserved source, it must couple universally like gravity. The theorem's *converse* does not hold: the existence of spin-2 tensor perturbations does not imply the existence of spin-2 particles.

In RS, curvature perturbations $h_{\mu\nu}$ of the effective metric satisfy the linearized Einstein equations in the continuum limit ($\lambda \gg \ell_0$):

$$\Box \bar{h}_{\mu\nu} = -2\kappa\, T_{\mu\nu}, \qquad \bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} h. \tag{6}$$

The tensor structure of $h_{\mu\nu}$ guarantees that free solutions have two polarization degrees of freedom ("+" and "×"), which is the hallmark of spin-2. But $h_{\mu\nu}$ is a perturbation of the ledger's effective metric, not a field with its own quantum excitations. The spin-2 character is kinematic (following from the symmetric rank-2 tensor nature of the metric) rather than dynamical (requiring a spin-2 particle).

The analogy is again precise: sound waves in a fluid exhibit spin-0 (longitudinal) polarization, but there is no fundamental "phonon particle" in the vacuum—phonons are emergent excitations of the lattice. Similarly, gravitational waves are emergent excitations of the ledger. □

6 Resolution of Non-Renormalizability

Theorem 6.1 (UV Finiteness of the Gravitational Sector)**.** *The non-renormalizability of perturbative quantum gravity does not arise in RS.*

Proof. The UV divergences of perturbative quantum gravity [3] arise from integrating graviton loop momenta to arbitrarily high energies: $\int d^4k\, k^{2n}$. In RS, this integral never appears for two independent reasons:

(i) No graviton, no graviton loops. Since there is no graviton field, there are no graviton propagators, no graviton vertices, and therefore no Feynman diagrams with graviton internal lines. The divergent integrals are artifacts of quantizing a field ($h_{\mu\nu}$) that does not exist as an independent degree of freedom.

(ii) The ledger provides a physical UV cutoff. Even for fields that *do* propagate on the ledger (the gauge fields of the Standard Model), the discrete lattice structure provides a natural UV cutoff at $k_{\max} = \pi/\ell_0$. Modes with wavelength shorter than $2\ell_0$ cannot propagate. This is not a regulator introduced by hand; it is a physical consequence of the ledger's discrete structure.

(iii) Gravity has no independent degrees of freedom. In RS, the gravitational "field" $g_{\mu\nu}$ is a derived quantity (the continuum limit of the ledger), not an independent field with its own action and equations of motion separate from the ledger. There are no gravitational degrees of freedom to renormalize. The matter fields live *on* the ledger and are subject to the lattice cutoff; the metric *is* the ledger and is not independently quantized. □

Remark 6.2. This should be contrasted with effective field theory (EFT) approaches [11], where gravity is treated as a well-defined quantum theory below the Planck scale, with higher-dimension operators suppressed by powers of M_{Pl}. In the EFT framework, "gravitons" are a useful computational device for computing one-loop corrections to, e.g., the Newtonian potential. RS does not dispute the numerical results of these calculations—the low-energy predictions of the EFT are reproduced by the continuum limit of the ledger. What RS disputes is the ontological interpretation: the EFT graviton is a mathematical convenience, not a physical particle.

7 The Effective Field Theory Perspective

A referee might ask: "If RS reproduces all low-energy gravitational predictions, including graviton-exchange calculations in the EFT framework, how can you distinguish between 'gravity is emergent' and 'gravity is mediated by gravitons'?"

Proposition 7.1 (Distinguishability at the Planck Scale)**.** *The RS prediction and the graviton hypothesis agree at energies $E \ll E_{\mathrm{Pl}}$ but diverge at $E \sim E_{\mathrm{Pl}}$.*

Proof. In the EFT framework, the graviton scattering amplitude at center-of-mass energy E is

$$\mathcal{M} \sim \frac{E^2}{M_{\mathrm{Pl}}^2}\left[1 + c_1\frac{E^2}{M_{\mathrm{Pl}}^2} + c_2\frac{E^4}{M_{\mathrm{Pl}}^4} + \cdots\right], \tag{7}$$

where the coefficients c_i encode UV physics. The EFT breaks down at $E \sim M_{\mathrm{Pl}}$, where all terms contribute equally. At this point, the EFT requires "UV completion"—a new theory (string theory, loop quantum gravity, etc.) that specifies the c_i.

In RS, there is no UV completion problem. The ledger lattice *is* the UV physics. The continuum limit reproduces the EFT at low energies, while the discrete structure resolves the Planck scale without introducing new particles or dimensions. The specific RS predictions at the Planck scale are:

(i) No scattering amplitude divergence: the lattice cutoff prevents $\mathcal{M} \to \infty$.

(ii) No new particle spectrum: RS introduces no tower of massive string states, no Kaluza–Klein modes, no superpartners.

(iii) Discrete spacetime: at distances $\sim \ell_0$, spacetime is a lattice, not a manifold.

□

8 Gravitational Waves Without Gravitons

Theorem 8.1 (Gravitational Wave Propagation)**.** *Gravitational waves are curvature perturbations of the ledger, propagating at exactly $c = \ell_0/\tau_0$.*

Proof. A local perturbation in the J-cost distribution at voxel $\mathbf{n}_0$ creates a curvature disturbance. The ledger's causal structure dictates that information propagates at the maximum signaling speed $c = \ell_0/\tau_0$ (one voxel per tick). Since both electromagnetic and gravitational perturbations propagate on the same ledger substrate with the same tick rate τ_0, both propagate at exactly c. There is no separate "gravitational medium" with a different propagation speed.

In the continuum limit ($\lambda \gg \ell_0$), the perturbation $h_{\mu\nu} \equiv g_{\mu\nu} - \eta_{\mu\nu}$ satisfies the linearized Einstein equation (6), yielding the standard wave equation $\Box h_{\mu\nu} = 0$ in vacuum. The dispersion relation $\omega^2 = c^2|\mathbf{k}|^2$ is exact, with no mass term and no frequency-dependent corrections. □

Corollary 8.2 (Graviton Mass Bound)**.** *The graviton mass is identically zero in RS, since there is no graviton. This is consistent with the LIGO–Virgo bound $m_g < 1.27 \times 10^{-23}$ eV$/c^2$ (GW170104 [12]) and the speed bound $|c_g/c - 1| < 10^{-15}$ (GW170817 [13]).*

Remark 8.3 (Energy and Angular Momentum). Gravitational waves carry energy and angular momentum—this is established observationally by the Hulse–Taylor binary pulsar decay [6] and directly by LIGO [5]. RS agrees: the GW energy is the kinetic energy of the curvature perturbation (the Isaacson effective stress-energy tensor), not the energy of a collection of graviton quanta. The distinction is analogous to ocean waves: they carry energy but are not composed of "ocean wave particles."

Theorem 8.4 (GR Recovery)**.** *In the continuum limit ($\lambda \gg \ell_0$), all GR predictions for gravitational wave generation, propagation, and detection are reproduced, including:*

(i) The quadrupole formula for gravitational radiation.

(ii) The chirp waveform for binary inspiral.

(iii) The ringdown quasi-normal modes determined by the Kerr metric.

(iv) The two tensor polarizations (+ and ×).

Proof. In the continuum limit, the emergent Einstein field equations (5) reduce to standard GR. All results that follow from the EFE—including the quadrupole formula, binary dynamics, and black hole perturbation theory—are therefore inherited by RS. The only regime where RS predictions differ from classical GR is at the Planck scale ($\lambda \sim \ell_0$), where the lattice structure becomes manifest. □

9 Comparison with Jacobson's Thermodynamic Gravity

The RS approach to emergent gravity has a noteworthy precursor in Jacobson's 1995 derivation of the Einstein field equations from the thermodynamics of local causal horizons [10]. Both approaches share the conclusion that gravity is not fundamental but emergent. However, they differ in the starting point:

Aspect	**Jacobson (1995)**	**RS**
Starting point	Clausius relation $\delta Q = T\,dS$	J-cost functional
Entropy	Bekenstein–Hawking $S = A/4$	J-cost density
Temperature	Unruh temperature	Recognition temperature
EFE status	Emergent (equation of state)	Emergent (action stationarity)
Graviton?	Not addressed	No (Theorem 4.1)
G	Free parameter	$G = \varphi^5$ (derived)

RS goes beyond Jacobson's result in two ways: it derives G (and hence κ) from the theory itself, and it provides the discrete substrate (the ledger lattice) from which the thermodynamic limit emerges.

10 Comparison with Quantum Gravity Approaches

Framework	**Graviton?**	**UV finite?**	**Bkgd. indep.?**	**G derived?**
Perturbative QG	Yes	No	No	No
String theory	Yes (closed)	Yes	Partially	No
Loop QG	No (spin-net.)	Yes	Yes	No
Asymp. safety	Yes (dressed)	Yes	Yes	No
Causal dyn. triang.	No	Yes	Yes	No
RS	No	Yes	Yes	Yes

RS is unique in deriving $G = \varphi^5$ from the cost functional, eliminating the gravitational coupling as a free parameter. Loop quantum gravity shares the "no fundamental graviton" feature but does not derive the numerical value of G.

11 The Bose–Marletto–Vedral Test

The Bose–Marletto–Vedral (BMV) experiment proposes to test whether gravity can create entanglement between two masses [8, 9]. The standard interpretation is: if entanglement is observed, then the gravitational field must have quantum degrees of freedom (i.e., gravitons).

Proposition 11.1 (RS Prediction for BMV)**.** *In RS, the BMV experiment* can *show gravity-mediated entanglement, but this does not require gravitons.*

Proof. In the BMV setup, two masses in spatial superposition interact gravitationally. If the gravitational interaction creates entanglement, standard arguments conclude that the mediator must be "quantum."

In RS, the mediator is the ledger itself, which already has discrete (quantum) structure. The two masses are ledger configurations; their gravitational interaction is mediated by their shared J-cost landscape. When each mass is in a spatial superposition, the ledger encodes all branches of the superposition (as distinct voxel configurations), and the gravitational interaction between branches creates correlations in the composite state.

The crucial distinction is:

(i) Standard QG: entanglement requires graviton exchange ($|g\rangle$ particles propagating between the masses).

(ii) RS: entanglement arises from the shared ledger structure, without any graviton particle.

Both frameworks predict the same observational outcome (entanglement), but they differ on the mechanism. RS predicts that the BMV entanglement rate is determined by $\kappa = 8\varphi^5 \approx 88.7$ (in RS-native units), not by a graviton exchange cross-section. □

Remark 11.2. If the BMV experiment finds *no* entanglement, this would falsify the standard graviton-exchange prediction. It would *not* falsify RS directly, because RS predicts entanglement through a different mechanism (shared ledger). However, the absence of entanglement would require RS to explain why the ledger correlations decohere in the experimental setup.

12 Falsifiability

Theorem 12.1 (Sharp Falsifier)**.** *Detection of a single graviton would falsify the no-graviton theorem.*

Remark 12.2. The graviton detection problem is practically unsolvable: the graviton interaction cross-section is $\sigma \sim GE^2/c^4 \sim 10^{-66}$ cm^2 at $E \sim 1$ GeV. Dyson has argued that no proposed experiment can detect individual gravitons [7]. Nevertheless, the prediction is sharp: if a graviton is ever detected, RS is falsified.

13 Additional Predictions

1. **No graviton detection**: All future experiments will fail to detect individual gravitons.

2. **Exact GW speed**: Gravitational waves propagate at exactly c at all frequencies. Any frequency-dependent dispersion $v(f) \neq c$ would indicate a massive graviton and falsify RS. Current bounds from GW170817 [13] constrain $|v_{\mathrm{GW}}/c - 1| < 10^{-15}$, consistent with RS.

3. **No graviton-induced decoherence**: Proposals to detect quantum gravity through decoherence from graviton emission [8] should find null results in the graviton-specific channel.

4. **Gravity-mediated entanglement via ledger**: The BMV experiment should observe entanglement, but the rate will be determined by $\kappa = 8\varphi^5$ (ledger mediation), not graviton exchange.

5. **No extra polarizations**: Gravitational waves have exactly two tensor polarizations ($+$ and $\times$). Detection of scalar or vector polarizations would indicate degrees of freedom beyond the metric tensor and would require modification of RS.

6. **No Planck-scale graviton spectrum**: Unlike string theory (which predicts a Regge tower of massive spin-J states), RS predicts no new gravitational particle spectrum at the Planck scale.

14 Conclusion

Gravitons do not exist in Recognition Science. Gravity is emergent curvature of the recognition ledger, arising from the J-cost functional through the Einstein coupling $\kappa = 8\varphi^5$. This resolves the non-renormalizability of perturbative quantum gravity, the background dependence problem, and the Weinberg spin-2 puzzle. Gravitational waves are curvature

perturbations of the ledger, not graviton beams; they exhibit spin-2 polarization as a kinematic consequence of the metric's tensor structure.

The prediction is sharply falsifiable: detection of a single graviton would refute RS. All GR predictions are recovered in the continuum limit, including the GW speed (exactly c), the quadrupole formula, and the binary inspiral waveform. RS is unique among approaches to quantum gravity in simultaneously denying the graviton, achieving UV finiteness, maintaining background independence, and deriving the numerical value of the gravitational constant $G = \varphi^5/\pi$.

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